A parity-calibrated equation in nine unknowns

Explicit Diophantine consistency reduction

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Abstract

An explicit selective source code, sparse signed-borrow padding, and a new parity calibration of the binary block give a polynomial of exact degree 7,344,328 in nine nonnegative or positive unknowns. Its positive form uses 738 binary arithmetic operations. Relative to a repository 129,875,980-degree, 788-operation nine-positive-variable comparator, the degree falls by a factor of about 17.68 and operations by 6.35%. The equivalence with absence of a finite ZFC refutation uses explicitly identified inherited compiler and Jones ratio theorems; no new proof of consistency or end-to-end kernel certificate is claimed.

1 Nine unknowns through a parity-calibrated carry block

The explicit polynomial below has nine nonnegative integer unknowns, exact degree 7 344 328, and 744 binary arithmetic operations. Its positive-coordinate translate has the same degree and 738 operations (211 additions, 26 subtractions, 501 multiplications). All intermediate names denote polynomial registers, not quantified variables. With the identified inherited compiler and Jones ratio theorems, ordinary mathematics formalizable in ZFC proves

$$\neg\exists \mathbf{x} \in \mathbb{N}^9 \; F_9(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is not a proof of consistency, a claim of minimum complexity, an unrestricted-integer nine-variable result, or a new end-to-end kernel proof.

1.1 Fixed data and selective extraction

The source is the original package31 system of 22 quadratic residuals in 31 positive coordinates, followed by the shift $y_i = x_i + 1$. Its exact JSON is `upstream/upstream_parent_quartic31.json`, SHA256 beginning `1c336ba0b3c9827f`; the full digest and repository pin `124c07c0b77a` are in `upstream/SOURCE_PINS.json`. We do not replace its transport row by the distinct package45 normalization. The whole 121-by-8801, 943-tile compiler tuple is bound in `COEFFICIENT_BINDING.json`; its four port bit lengths are $(1064921, 1070004, 1064921, 1064922)$. The M_F port is the already shifted $M_{F,\mathrm{source}}$. All four authenticated compressed little-endian payloads and the complete tuple adapter proof are included.

For the shifted residual p_j, set

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, [\mathbf{x}^\alpha] p_j.$$

Exact source expansion produces 136 terms and 26 distinct recipes, including literal-only recipes. From the port bit lengths, termwise triangle bounds prove $|c_{j,\alpha}| < M = 2^{2134927}$. Fix

$$\kappa = 2134940, \qquad a = 2^\kappa = 8192M, \qquad c = 2^{15\kappa/2-10} = 2^{16012040}.$$

These are fixed ordinary integers. The evenness of κ will matter.

The 31 weights, in the source variable order, are

$$9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7, 135,$$
$$38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120.$$

The row order and corresponding targets are

$$16, 2, 9, 14, 21, 7, 15, 22, 13, 3, 19, 20, 1, 10, 17, 8, 4, 6, 11, 18, 12, 5;$$
$$70, 86, 451, 685, 1050, 1296, 1530, 1874, 2212, 2358, 2696, 2780,$$
$$2792, 3157, 3290, 3634, 3929, 4238, 4384, 4527, 4865, 5190.$$

Write $W = 183$, $Q_c = 5190$, $K_c = Q_c + 2W = 5556$, and $E_c = K_c + W + 2 = 5741$. The frozen finite certificate verifies, for every source-supported α and every $|\beta| \le 2$,

$$q_j - \alpha \cdot w + \beta \cdot w = q_l \quad \Longleftrightarrow \quad j = l \text{ and } \alpha = \beta.$$

It checks all $136 \cdot 528 = 71808$ interactions. Only the 71 used padded pairs need isolation; absent pair sums may coincide. The reversed exponents are nonnegative and mutually distinct, so no coefficient inflation factor occurs. Define

$$D_c(t) = \sum_{j,\alpha} c_{j,\alpha} t^{q_j - \alpha \cdot w}, \qquad C_{\mathbf{x}}(t) = \left(1 + \sum_i x_i t^{w_i}\right)^2.$$

Factorial cancellation gives $[t^{q_j}](C_{\mathbf{x}} D_c) = 2p_j(\mathbf{x})$. The highest coefficient of D_c is 2 at exponent 5190: residual5 is last and has constant term 1. Thus, for $B - 1 > M$,

$$B^{5190} < D_c(B) < \frac{MB^{5191}}{B-1}.$$

1.2 Sparse target padding and source bounds

Use $(\epsilon, g_0, h_0, s_0, w_0, \phi, i_0, j_0, n) \in \mathbb{N}^9$, where $\mathbb{N} = \{0, 1, \ldots\}$. Put $g = g_0 + 1, h = h_0 + 1, s = s_0 + 1, w = w_0 + 1, i = i_0 + 1, j = j_0 + 1$, and define

$$b = \epsilon + 2, \qquad B = ab^2, \qquad \ell = \sum_i B^{w_i}, \quad A_t = \sum_j B^{q_j},$$
$$N_0 = B^{184}, \qquad Z = B^{5557}, \qquad L = 2bB^{5741}, \quad N = cB^{2966},$$
$$M_c = N_0 - 1 - (b-1)\ell, \qquad J_c = 2(1+g)^2 D_c(B) + A_t,$$
$$S_c = g + N_0 J_c, \qquad T_c = M_c + N_0(B-2)A_t,$$
$$R = L^2 + (S_c + T_c)L + L - 1 - S_c, \quad G_{\text{cap}} = 32bB^{183}.$$

The block length L and ratio threshold root N are different. On every outer assignment, $J_c, S_c > 0$, and the base-B digits of M_c are $B - b$ at the weights and $B - 1$ elsewhere through W. Hence $0 \le M_c < N_0$, $T_c \ge 0$, and

$$R = L^2 + (L-1)S_c + LT_c + L - 1 \ge 8, \qquad 0 < b \le R, N, \quad N \ge 8, \quad G_{\text{cap}} > 0.$$

These facts do not presuppose a power-of-two base or a bound on g.

If $g < 4bB^{183}$, integrality yields $1 + g \le 4bB^{183}$ and $g < N_0$. Therefore

$$2(1+g)^2 D_c(B) < \frac{32Mb^2}{B-1} Z = \frac{B}{256(B-1)} Z < \frac{Z}{128}, \qquad A_t < \frac{Z}{128}.$$

Thus $0 < J_c < Z/64 < Z$, while $(B-2)A_t < B^{5191} \le Z$. Using integrality again gives

$$0 \le S_c, T_c < N_0 Z = B^{5741}, \qquad S_c + T_c < 2B^{5741} < L.$$

Once b is a power of two, so is B. Binary disjointness of g and M_c is exactly $g = \sum_i x_i B^{w_i}$ with $0 \le x_i < b$. Let $h_r = [t^r](C_{\mathbf{x}} D_c)$. Then

$$|h_r| < M(1 + \sum x_i)^2 < 1024 M b^2 = B/8, \qquad J_c = \sum_{r=0}^{5556} (2h_r + \delta_r) B^r,$$

where δ_r is 1 at a target and 0 elsewhere. In signed-digit normalization the incoming borrow is always -1 or 0: the coefficient bound preserves this by induction. At a target the normalized digit is $(2h_r + 1 + \text{borrow}) \bmod B$. It avoids the mask $B-2$ precisely when $h_r = 0$: for zero it is 0 or 1; for positive h_r it is at least 2 and below $B/2$; for negative h_r it is above $B/2$. Consequently

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g \text{ codes a common nonnegative zero of all } p_j.$$

The symbol & denotes ordinary bitwise intersection of nonnegative integers. It occurs only in the proof, not in the polynomial circuit.

1.3 The new block-parity calibration

For $S, T \ge 0$, $S + T < L = 2^u$, the polynomial

$$X = L^2 + (S+T)L + (L - 1 - S)$$

has disjoint top, middle and bottom binary blocks. Thus

$$\operatorname{pop}(X) = u + \operatorname{pop}(T) + 1 - \big[\operatorname{pop}(S) + \operatorname{pop}(T) - \operatorname{pop}(S+T)\big].$$

The bracket is nonnegative and vanishes exactly if $S \,\&\, T = 0$. Legendre's formula gives $v_2\binom{2X}{X} = \operatorname{pop}(X)$, so

$$2^{u + \operatorname{pop}(T) + 1} \mid \binom{2X}{X} \quad \Longleftrightarrow \quad S \,\&\, T = 0.$$

Put $b = 2^k, B = 2^v$, with $v = \kappa + 2k$. The low mask contributes $184v - 31k$ bits; the 22 high target digits contribute $22(v-1)$. Hence

$$\operatorname{pop}(T_c) = 206v - 31k - 22.$$

Choosing $L = 2bB^{5741}$ gives $u = 1 + k + 5741v$, and therefore

$$\begin{aligned} u + \operatorname{pop}(T_c) + 1 &= 5947v - 30k - 20 \\ &= 5947\kappa + 11864k - 20 \\ &= 2\big(15\kappa/2 - 10 + 2966v\big) = 2\log_2 N. \end{aligned}$$

Thus the tested threshold is exactly N^2, even with b linear in the input. No extra unknown, dummy digit or square-base restriction is needed. This is the new parity-calibration step.

Conversely, for a source zero choose any sufficiently large power of two $b > \max x_i$, and $\epsilon = b - 2$. Since $p_5(0) = 1$, its code g is positive. Moreover $g \le (b-1)\ell < bB^{183}$, so $32g < G_{\text{cap}}$. Both masks are disjoint and $N^2 \mid \binom{2R}{R}$. Every sufficiently large power of two is available, eliminating any hidden cofinality assumption.

1.4 Ratio registers and the near-square finalizer

Use the inherited Jones ratio registers

$$U = N^2 w, \quad Y = N^2 s, \quad M_0 = RY, \quad P = 2M_0^2 U,$$
$$K = R + 1 + h(P-1), \quad A = M_0(U+1), \quad B_1 = 2R+1, \quad W_1 = bw,$$
$$C = B_1 + W_1 + \phi, \quad J = 4A - 5, \quad Q = 3W_1 C - 2(W_1^2 - 1),$$
$$D = (A^2-1)C^2 + 1, \quad E = iJDC^2, \quad F = (A^2-1)E^2 + 1,$$
$$G = A + F(CD + 1 - A), \quad H = B_1 + jC, \quad I = (G^2-1)H^2 + 1,$$
$$a_0 = (P^2-1)K^2 + 1, \quad a_1 = 64DFI, \quad u_0 = JF, \quad v_0 = QF + J(H-C),$$
$$\mu = K^2 G_{\mathrm{cap}} - 4(C - KY)^2 G_{\mathrm{cap}} - 8gK^2.$$

Here u_0, v_0 avoid conflict with the binary block exponent. The imported `ratio_sound` and `ratio_complete` in `upstream/NineRatio.lean` relate the four conditions

$$a_0 \text{ square}, \quad DFI \text{ square}, \quad u_0 \mid v_0, \quad \mu > 0$$

to the central-binomial divisibility and b a power of two. Soundness gives $8g < G_{\mathrm{cap}}$; completeness requires $32g < G_{\mathrm{cap}}$. All numerical hypotheses were proved unconditionally above. Multiplying DFI by 64 does not change its square predicate.

For every outer assignment, $A \geq 2, J \geq 3, D \geq A^2, K > 1$, all Pell registers are positive, and $Q, H - C > 0$. Thus

$$0 < \frac{v_0}{u_0} = \frac{Q}{J} + \frac{H-C}{F} \leq Q + H - C =: M_d.$$

Put

$$V_0 = PK, \quad t_0 = iJC, \quad S_0 = t_0(8D^2 - 4D - 1), \quad V_1 = S_0 GH.$$

As $E = t_0 DC$, exact identities yield

$$DF = t_0^2 D^3(D-1) + D, \qquad S_0^2 - 64DF = t_0^2(8D+1) - 64D \geq 8D + 9 > 0.$$

Since $I \leq G^2 H^2$ and $V_0^2 - a_0 = K^2 - 1 > 0$, we have $0 < a_r < V_r^2$ for $r = 0, 1$. Define the requested single polynomial by

$$\Omega = u_0(2\mu - 1), \qquad \Theta = u_0 n + v_0 - \Omega(M_d + V_0 V_1),$$
$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1.}$$

The four-factor identity implies, at any zero, $\Theta/\Omega = r_0 + V_0 r_1$ with $r_i^2 = a_i$. This rational number cannot be zero, because then $|r_1| = |r_0|/V_0 < 1$, contrary to the positive integer a_1. The identity $2(\Theta/\Omega)r_0 = (\Theta/\Omega)^2 + a_0 - V_0^2 a_1$ forces both roots rational and thus integral. This argument does not assume independence of square classes. The strict bounds give

$$Z_0 = M_d + V_0 V_1 + r_0 + V_0 r_1 \geq M_d + 1 > 0.$$

The factor equation is $n + v_0/u_0 = (2\mu - 1)Z_0$. Its positive left side forces $\mu > 0$, and its integral right side forces $u_0 \mid v_0$. Conversely the four ratio conditions yield an integer $n = (2\mu - 1)Z_0 - v_0/u_0 \geq 1$ and hence a zero. This completes both directions of the stated reduction.

1.5 Exact degree and audited implementation

A raw degree walk would miss a leading cancellation. The certificate uses

$$\begin{aligned}\Delta_V = V_1^2 - a_1 &= [t_0^2(8D+1) - 64D]G^2H^2 + 64DF(H^2-1),\\ E_0 &= u_0 n + v_0 - \Omega M_d,\\ \Theta^2 - \Omega^2(a_0 + V_0^2 a_1) &= E_0^2 - 2E_0\Omega V_0 V_1 + \Omega^2 V_0^2 \Delta_V - \Omega^2 a_0.\end{aligned}$$

The first summand of Δ_V has degree $1\,763\,760$ and leader $2^{147}a^{881800}c^{96}$. The third summand of the last display dominates, with degree $3\,672\,164$ and leader $2^{299}a^{1835914}c^{224}$. Finally $4\Omega^4 V_0^2 a_0 a_1$ has degree $5\,998\,540$, strictly less than twice the dominant degree. Thus

$$\deg F_9 = 7\,344\,328, \qquad \mathrm{LC}_{\mathrm{diagonal}}(F_9) = 2^{598}a^{3671828}c^{448} > 0.$$

The diagonal attains the full multivariate total-degree bound. The cancellation-resolved certificate records every decisive exact leader; coefficient-dependent lower heads are left unknown, not guessed.

Replacing every natural input by a positive input minus one is a bijection and preserves degree. In particular $b = \epsilon_+ + 1$, $\phi = \phi_+ - 1$, and $n = n_+ - 1$. The conservative three-block comparison has degree $12\,023\,320$ and 734 positive operations. A square-base uncalibrated two-block comparison has degree $14\,687\,132$ and 740 operations. Its admissible bases $b = 2^{2r+1}$ are explicitly cofinal. Both comparison circuits and their proof details are included; the parity-calibrated version is primary.

The self-contained standard-library scripts `build_nine.py` and `verify_nine.py` rebuild all 12 circuits, authenticate the full tuple, verify all 26 recipe bounds and the selective certificate, and perform 360 modular replays covering 11520 independently computed named registers. They also check 43945 two-block cases, 42 threshold identities, an exact symbolic near-square identity, 243 majorant fixtures and 1728 finalizer fixtures. These finite checks audit data and implementation, not consistency. The complete proof and all limitations are in `PROOF.md`; provenance and imported dependencies are distinguished from the new mathematics.